Medicines & antibodies

Treat disease

1 Antibiotics & symptoms

Antibiotics such as penicillin treat bacteria; choose an effective one. **Resistance** makes infections harder to treat.

Antibiotics do not kill viruses. Viruses reproduce inside cells, making safe targeting difficult.

Painkillers relieve symptoms without killing pathogens.

2 Drug origins

Digitalis ← foxglove. Aspirin ← willow.

Penicillin: Fleming's discovery from Penicillium mould.

Most new drugs are synthesised; natural chemicals may provide starting points.

Test new drugs

3 Testing sequence

Check **toxicity, efficacy and dose.**
Preclinical: cells, tissues, live animals.

Clinical: low doses in healthy volunteers, then patients to test effectiveness and optimum dose.

Results undergo **peer review** before publication.

4 Fair comparisons

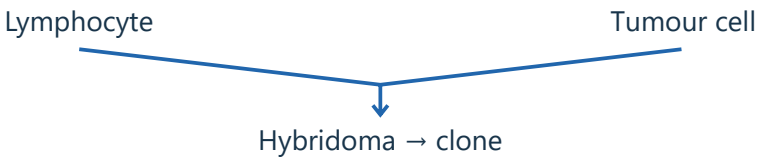
Placebo: no active drug. **Double-blind:** patients and assessors do not know allocations.

Random allocation, comparable groups and large samples reduce misleading effects.

Compare recovery rates and side effects, not raw counts alone.

Target one antigen

5 Make monoclonal antibodies



Antigen stimulates mouse lymphocytes. Fuse an antibody-producinglymphocyte with a tumour cell → **hybridoma.**

Select and clone one hybridoma; collect/purify antibodies. Clone makes antibodies with **one specific binding site.**

6 Uses & limits

Specific binding: pregnancy tests, hormone measurement, pathogen detection; fluorescent tags locate molecules.

Attach toxic drugs/radioactive substances/growth blockers to target cancer cells.

Targeting reduces unwanted damage but side effects and unsuitable targets can limit benefit.